Regime Detection using Sliced Wasserstein k-means clustering with Monte Carlo

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1 Abstract

This thesis investigates the topic of Regime Detection using an interpretable Machine Learning algorithm. In particular, the focus of this thesis is to produce a tractable algorithm for detecting regimes in financial markets based on multi-dimensional time series. This thesis, provides an enhanced algorithm to cluster the multi-dimensional returns distributions. Depending, on the the characteristics of the distributions, the algorithm can infer which regime the market is in without pre-specified assumptions on the returns distribution. Specifically, it uses the Wasserstein distance as a metric to compare different regimes. In higher-dimensional settings, one can extend this concept using the Sliced Wasserstein distance and use an algorithm based on this metric to cluster different market regimes (Sliced Wasserstein k-means clustering algorithm). The following thesis is structured as follows believed to be the most explanatory and logical way to present the research.

2 Experiment 1: Theoretical Framework for Wasserstein Distance/Sliced Wasserstein (SW) Distance

This chapter will focus on the theoretical framework to understand the Wasserstein distance and its extension to the Sliced Wasserstein distance. Moreover, this section will provide insights on the practical implementation of the above-mentionned algorithms and limitations to the current SW clustering implementations in the literature. Notably, the low generalizable nature of using pre-defined projection vectors θ^l for the Sliced Wasserstein can be solved by generating them using the UnifOrtho method [4]. It provides the following advantages compared to standard Monte Carlo methods.

- **Variance reduction via Negative Negative Dependance**
- **Superior performance in high dimensions** : The UnifOrtho is therefore perfectly adapted to working with multi-dimensional time series for the SW k-means clustering algorithm.
- **Clustering Stability**

Some useful theoretical results on variance reduction and computational cost on the UnifOrtho method will therefore be provided to support the choice of this method for the future sections.

The following pseudocode is based on the work by [4] on Repulsive Monte Carlo Methods and [3] on Sliced Wasserstein k-means clustering for regime detection.

Algorithm 1: sWk-means clustering with UnifOrtho Projections

Input: Data stream $S \in \mathcal{S}(\mathbb{R}^d)$,;
number of clusters K,;
number of projections L,;
tolerance ϵ for convergence
Output: K cluster assignments and corresponding 1d barycenters

▷ `Preprocessing`
Calculate $l(r^S)$ given $S \in \mathcal{S}(\mathbb{R}^d)$ and define family of empirical distributions $\mathcal{K} = \{\mu_j\}_{1 \leq j \leq M}$;
▷ `Step 1: UnifOrtho Projection Generation`
$k \leftarrow \lceil L/d \rceil$; ▷ `number of orthogonal bases needed`
$\Theta \leftarrow \emptyset$;
for $i = 1$ **to** k **do**
  Sample a random matrix $Z_i \in \mathbb{R}^{d \times d}$ with i.i.d. $\mathcal{N}(0, 1)$ entries;
  Compute the QR decomposition: $Z_i = Q_i R_i$;
  Let Λ_i be a diagonal matrix where $\Lambda_{i,jj} = \mathrm{sgn}(R_{i,jj})$;
  $U_i \leftarrow Q_i \Lambda_i$; ▷ U_i `is Haar-distributed on` $O(d)$
  Add the d column vectors of U_i to Θ;
$\{\theta^1, \ldots, \theta^L\} \leftarrow \Theta[1 : L]$; ▷ `select exactly` L `directions`
▷ `Step 2: Projection & K-Means Iteration`
Calculate projected distributions $\{\mu'_j(\theta^l)\}_{1 \leq j \leq M}$ for all $l = 1 \ldots L$;
Initialize cluster assignments $C_1, \ldots, C_K$;
while *convergence criterion* $> \epsilon$ **do**
  for $k = 1$ **to** K **do**
    Compute the 1d Wasserstein barycenter $\bar{\mu}_k(\theta^l)$ for each projection l using empirical distributions assigned to C_k;
  for $j = 1$ **to** M **do**
    Assign μ_j to the cluster C_{k^*} where $k^* = \arg\min_k \sum_{l=1}^{L} W_p^p\big(\mu'_j(\theta^l), \bar{\mu}_k(\theta^l)\big)$;
return $C_1, \ldots, C_K$ *and barycenters* $\{\bar{\mu}_k(\theta^l)\}$

3 Experiment 2: Design methodology for Monte Carlo for SW - Synthetic Data Testing

The focus of this chapter will be on testing the algorithm on syntethic data to understand the behaviour of the algorithm and its performance in different settings. The algorithm will be tested on 3 dimensional data followed by 4, 5 dimensional. Performance between original and UniOrtho method will be compared and tests will span from data with multiple correlation factors and with diffrent target number of clusters. Finally, in terms of time-horizon, the study will focus on the equivalent of 20 years synthetic data with 7 observations a day (252 trading days a year) to be as close as possible to real world data.

Different performance metrics will be used for the regime detection :

- Total Accuracy (TA)
- Mean centroid-centroid distance (MCCD)
- Point-to-centroid distance (PCD)

The synthetic data will be generated similarly to previous papers [1] and [3] using Geometric Brownian Motion (GBM) paths $S_t/S_{t-1} = \exp(r_t^S)$ with $S_t = (S_t^1, \ldots, S_t^d)$ and $r_t^S = (r_t^{S^{(1)}}, \ldots, r_t^{S^{(d)}})$ where :

$$r_t^{S^{(i)}} \sim \mathcal{N}\left((\mu - \sigma^2/2)\mathbf{1}dt, \mathbf{\Sigma}dt\right) \tag{1}$$

. with

$$\Theta = (\mu, \sigma^2) \tag{2}$$

And the covariance matrix $\mathbf{\Sigma}$ is defined as follows :

$$\Sigma_{ij} = \sigma^2 \left(\delta_{ij} + (1 - \delta_{ij})\rho\right) \tag{3}$$

The δ_{ij} is the Kronecker delta, σ^2 is the variance of the returns and ρ is the correlation between the different assets.

Assuming, we want to cluster two different regimes (K=2), we define, as in previous papers [3], Bull and Bear regimes with the following parametrization :

- $\Theta_{\text{bull}} = (0.02, 0.2)$
- $\Theta_{\text{bear}} = (-0.02, 0.3)$

For K = 3, similarly to [3] we will consider the *moon-shape structure* generated using `datasets.make_moons` function from the `sklearn` library.

This is a summary similary to the one provided in [3] for the different market settings on which we are going to test the sWk-means clustering algorithm with UnifOrtho projections :

Table 1: Summary of synthetic dataset parameters with $d \in \{3, 4, 5\}$

Type	Regime Bull	Regime Bear	Regime III (Bear/Moon)
Type A (K=2)	$\Theta_{\text{bull}}; \rho = +1/2$	$\Theta_{\text{bear}}; \rho = +1/2$	N/A
Type B (K=2)	$\Theta_{\text{bull}}; \rho = +1/2$	$\Theta_{\text{bull}}; \rho = -1/2$	N/A
Type C (K=3)	$\Theta_{\text{bull}}; \rho = +1/2$	$\Theta_{\text{bear}}; \rho = +1/2$	$\Theta_{\text{bear}}; \rho = -1/2$
Type D (K=3)	$\Theta_{\text{bull}}; \rho = +1/2$	$\Theta_{\text{bear}}; \rho = +1/2$	$\Theta_{\text{moon}}; \rho = +1/2$

4 Experiment 3: Application of Monte Carlo SW for Real Data and Analysis

Finally, the chapter will focus on the application of the novel algorithm on real world data sets for different types of financial markets. As above, the time span, wil focus on 15 to 20 years of data. The target data sets include the following :

- FX Spot Series Data (4D)

  EURUSD, AUDUSD, USDJPY, USDCHF, USDCAD (Major Currency Cluster) (15 min data)

- Equity Series Data (3D)

  S&P 500, NASDAQ, Dow Jones (US Equity Cluster) (daily data)

  CAC 40, DAX40, FTSE 100 (European Equity Cluster) (daily data)

  Nikkei 225, Hang Seng, Kospi 200 (Asian Equity Cluster) (daily data)

- Commodity Series Data (4D and 5D)

  WTI Crude Oil, Brent Crude Oil, Natural Gas, Heating Oil.

  Copper Aluminum, Zinc, Nickel, Lead.

5 Experiment 4: Performance analysis of regime detection using trading strategies

- Long/Short strategy based on majority regime detection signals
- Long/Short strategy based on Implied probability signals (using the distance to the centroids as a proxy for the confidence of the regime detection)
- Ensemble Methods: combining the above two strategies using a weighted average of the signals or an adaptative weighting scheme to generate the final trading signal.

Tested trading strategies on FX and Equity data sets. Performance is compared to a long only and short-only approach. Moreover, for equities, we look at the performance for different clusters (US, Europe, Asia) to see if the performance of the regime detection is better in some markets than others. We also test, trading algorithm that use american regime detection to trade in Asian and European markets to see if the regime detection is more generalizable in some markets than others.

5.0.1 Standard SW k-means trading strategy

Pseudo code for this strategy is provided here :

Algorithm 2: LONG_STRAT_UNIFORTHO — Binary Regime Strategy

Input: Initial capital C_0,;
number of simulations N_S,;
price matrix $S \in \mathbb{R}^{T \times d}$,;
trimming parameters h_1, h_2,;
window size w,;
number of clusters K,;
risk metric m,;
majority look-back ℓ
Output: Portfolio value path V

$R \leftarrow \text{PctReturns}(S)$; ▷ $R_t = S_t / S_{t-1} - 1$
$\theta \leftarrow \mathbf{1}_d / d$; ▷ `equal-weight vector`
$r^{(\text{pf})} \leftarrow R\,\theta$; ▷ `portfolio return series`
$V \leftarrow [C_0]$;
$n \leftarrow \lfloor T/w \rfloor$;
for $i = 0$ **to** $n-2$ **do**
  $s \leftarrow i \cdot w, \quad e \leftarrow (i+1) \cdot w$;
  $W \leftarrow S[s:e]$; ▷ `current analysis window`
  ▷ `Step 1 -- Wasserstein-based regime detection`
  $\hat{F},\ \boldsymbol{\mu},\ \boldsymbol{\lambda} \leftarrow$ MAXMCCD_UNIFORTHO$(N_S,\ W,\ K,\ L,\ \varepsilon,\ h_1,\ h_2,\ m)$;
  ▷ `Step 2 -- Majority-vote regime classification`
  if $\ell > |\boldsymbol{\lambda}|$ **then**
    $\hat{k} \leftarrow \arg\max_k\ \text{count}(\boldsymbol{\lambda} = k)$;
  else
    $\hat{k} \leftarrow \arg\max_k\ \text{count}(\boldsymbol{\lambda}_{[-\ell:]} = k)$;
  ▷ `Step 3 -- Generate binary signal`
  if $\hat{k} = 1$ **then**
    $\sigma \leftarrow +1$; ▷ `Bullish` $\Rightarrow$ `Long`
  else
    $\sigma \leftarrow -1$; ▷ `Bearish` $\Rightarrow$ `Short`
  ▷ `Step 4 -- Compound portfolio over next window`
  for *each* $r_t \in r^{(\text{pf})}[e : e+w]$ **do**
    $V.\text{append}\Big(V[-1]\ \times\ \big(1 + \sigma\, r_t\big)\Big)$;
return V

5.0.2 Implied Probability SW k-means trading strategy

Theoretical Background We use implied probability measure to compute wether on thinks the market is going to become bear or bull.

Likelihood Function It uses SW-distance to every centroid to every centroid and then convert the likelihood via a softmax over the negative distances. We introduce a temperature parameter τ

$$\mathcal{L}(\text{regime } k \mid x_m) = \frac{e^{-d(x_m, c_k)/\tau}}{\sum_j e^{-d(x_m, c_j)/\tau}}$$

Empirical Transition Matrix Markov Prior A market in regime i is highly likely to stay in regime i tomorrow. We model this as a first-order Markov Chain using the historical labels. Probability of switching from i to j is:

$$\pi_{i,j} = P(Z_m = j \mid Z_{m-1} = i) = \frac{\text{Count}(i \to j)}{\sum_{x=1}^{K} \text{Count}(i \to x)}$$

If the market was $z_{current}$ our prior belief that state M is going to be at regime k is :

$$\mathbb{P}_{prior}(k) = \pi_{z_{current},k}$$

Then we simply use Bayes rule to derive the following :

$$\mathbb{P}_{Bayes}(k) = \frac{\mathbb{P}_{prior}(k) \cdot \mathcal{L}(k \mid X_m)}{\sum_{j=1}^{K} \mathbb{P}_{prior}(j) \cdot \mathcal{L}(j \mid X_m)}$$

Switch Probability Finally the switch probability does the following :

$$\mathbb{P}_{switch} = 1 - \mathbb{P}_{Bayes}(z_{current})$$

Use Gradient One may also want to take into account the fact that one is moving away or getting closer to a certain regime.

We consider a lookback window of size ω, and one computes the linear trend slope of the Wasserstein distance to each centroid k. Let $t \in \{1, 2, \ldots, \omega\}$ We get using OLS, β_k:

$$\beta_k = \frac{\text{Covariance}(t,\, \mathcal{W}_2(X_{M-w+t},\, C_k))}{\text{Variance}(t)}$$

- if $\beta_k < 0$ the distance to regime k is shrinking (we are getting close to regime k)
- if $\beta_k > 0$ the distance to regime k is growing (we are drifting away from regime k)

We apply softmax to get a probability:

$$P_{Grad}(k) = \frac{\exp\left(-\frac{\beta_k}{\tau}\right)}{\sum_{j=1}^{K} \exp\left(-\frac{\beta_j}{\tau}\right)}$$

The final switch probability using gradient is therefore:

$$\mathbb{P}_{switch} = 1 - [(1-\alpha)\mathbb{P}_{Bayes}(z_{current}) + \alpha \mathbb{P}_{Grad}(z_{current})]$$

with $\alpha \in [0 : 1]$ a weight.

Algorithm 3: LONG_STRAT_IMPLIED — Implied-Probability Regime Strategy

Input: Initial capital C_0,;
simulations N_S,;
price matrix $S \in \mathbb{R}^{T \times d}$,;
trimming h_1, h_2,;
window w,;
clusters K,;
metric m,;
signal type $\tau \in \{$`continuous`, `hysteresis`, `conviction`$\}$,;
entry threshold α_e,;
hold threshold α_h,;
look-back ℓ,;
gradient flag and weight (g, ω_g)
Output: Portfolio value path V,;
signal history Σ

$R \leftarrow \text{PctReturns}(S), \quad \theta \leftarrow \mathbf{1}_d/d, \quad r^{(\text{pf})} \leftarrow R\,\theta$;
$V \leftarrow [C_0], \quad \Sigma \leftarrow [\,], \quad n \leftarrow \lfloor T/w \rfloor$;
for $i = 0$ **to** $n - 2$ **do**
  $s \leftarrow i \cdot w, \quad e \leftarrow (i+1) \cdot w, \quad W \leftarrow S[s:e]$;
  `▷ Step 1 -- Regime detection (same as Algorithm 2)`
  $\hat{F},\ \boldsymbol{\mu},\ \boldsymbol{\lambda} \leftarrow$ MAXMCCD_UNIFORTHO(N_S, W, K, L, ε, h_1, h_2, m);
  `▷ Step 2 -- Implied transition probabilities`
  $\mathbf{P},\ p_{\text{switch}},\ \mathbf{T},\ \boldsymbol{\pi} \leftarrow$ COMPUTEIMPLIEDPROBA($\hat{F}$, $\boldsymbol{\mu}$, $\boldsymbol{\lambda}$, ℓ, g, ω_g);
  $\hat{k} \leftarrow \arg\max_k\ \text{count}(\boldsymbol{\lambda}_{[-h_2:]} = k)$;
  `▷ Step 3 -- Signal generation (depends on τ)`
  if $\tau =$ *continuous* **then**
    $\sigma \leftarrow \pi_{\text{bull}} - \pi_{\text{bear}}$;
    if $|\sigma| < \delta$ **then**
      $\sigma \leftarrow 0$
    ; `▷ dead zone`
  else if $\tau =$ *hysteresis* **then**
    $\sigma_{\text{prev}} \leftarrow \Sigma[-1]$ **or** 0;
    if $\hat{k} = \textit{Bull}$ **then**
      if $p_{switch} \geq \alpha_e$ ***and*** $\sigma_{prev} \geq 0$ **then** $\sigma \leftarrow -1$;
      else if $p_{switch} < \alpha_h$ ***and*** $\sigma_{prev} < 0$ **then** $\sigma \leftarrow +1$;
      else $\sigma \leftarrow 0$;
    else `▷ k̂ = Bear`
      if $p_{switch} \geq \alpha_e$ ***and*** $\sigma_{prev} \leq 0$ **then** $\sigma \leftarrow +1$;
      else if $p_{switch} < \alpha_h$ ***and*** $\sigma_{prev} > 0$ **then** $\sigma \leftarrow -1$;
      else $\sigma \leftarrow 0$;
  else if $\tau =$ *conviction* **then**
    $d_k \leftarrow \begin{cases} +1 & \hat{k} = \text{Bull} \\ -1 & \hat{k} = \text{Bear} \end{cases}$;
    $c \leftarrow 1 - 1.5\,p_{\text{switch}}$; `▷ conviction decays with switch risk`
    $\sigma \leftarrow d_k \cdot c$;
  Σ.append(σ);
  `▷ Step 4 -- Compound portfolio over next window`
  for *each* $r_t \in r^{(\text{pf})}[e : e + w]$ **do**
    $V.\text{append}\Big(V[-1]\ \times\ \big(1 + \sigma\, r_t\big)\Big)$;
return V, Σ

5.1 Performance Analysis for Trading Strategies: Ledoit-Wolf test on sharpe ratios

Conducted Ledoit-Wolf test to compare the Sharpe ratios of the different trading strategies and see if the differences in performance are statistically significant.

Need to read more about shap analysis in [2]

6 Experiment 5: Interpretability Analysis of Sliced Wasserstein Regime Detection Algorithm

We label the regimes using weekly equity data (S&P 500, NASDAQ, Dow Jones) on the period 2006 to 2020. We further consider the following macro feature set:

- VIX Index
- 10-year Treasury Yield
- 5-year Treasury Yield
- 30-year Treasury Yield
- Fed Funds Rate
- Manufacturing PMI
- Manufacturing Employment
- DXY Dollar Index
- Gold Spot Price
- Brent Crude Oil Spot Price

After having regrouped the data from the following macro-indicators from the appropriate time period, we fit thefeature set to the regime labels using the following models:

- OLS Linear Regression (probabilistic target (Probability of being in bear regime))
- Logistic Regression (binary target (Bull/Bear))
- Random Forest Classifier (binary target (Bull/Bear))

In particular, we are analysing the feature importance of the different macro-indicators during the bear regimes using the SHAP analysis toolbox. We observe that in particular during predicted bear regimes by the sliced-Wasserstein k-means clustering algorithm, the VIX index is a particularly important feature in the regime classification for all three models.

7 Appendix

References

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